Skin Lesion Detection Using YOLOv8

SOC FINAL TERM SUBMISSION REPORT

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Project: AI for Healthcare

1. Introduction

Skin lesions are abnormal growths or changes in the skin that can range from harmless moles to early signs of skin cancer such as melanoma. Early and accurate identification of suspicious lesions greatly improves the chances of successful treatment, since skin cancers detected at an early stage are far easier to manage than those diagnosed later.

Dermoscopic and clinical photography are widely used to examine skin lesions because they provide a clear, magnified view of the lesion's structure, colour, and border characteristics. However, manual screening of large volumes of skin images is time-consuming and depends heavily on the expertise of the examining dermatologist.

In recent years, deep learning has shown remarkable success in medical image analysis. Among different object detection algorithms, YOLO (You Only Look Once) has become popular because of its speed and accuracy.

In this project, YOLOv8 was used to automatically detect skin lesions in dermoscopic/clinical images by localizing the lesion using bounding boxes.

2. Objectives

The objectives of this project were:

- Detect skin lesions from dermoscopic/clinical images automatically.
- Localize lesions using bounding boxes.
- Train a YOLOv8 object detection model.
- Evaluate the model using standard performance metrics.
- Analyze prediction accuracy on unseen skin images.
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3. Dataset

The dataset used in this project was obtained from Roboflow Universe.

Dataset Details

- Dataset: Skin Lesion Detection
- Image Type: Dermoscopic / clinical skin images
- Number of Classes: 1 (Skin Lesion)
- Total Annotated Lesion Instances (training set): approximately 1,490

Each image contains manually annotated bounding boxes indicating the lesion location.

The dataset was divided into:

- Training Set
- Validation Set
- Testing Set

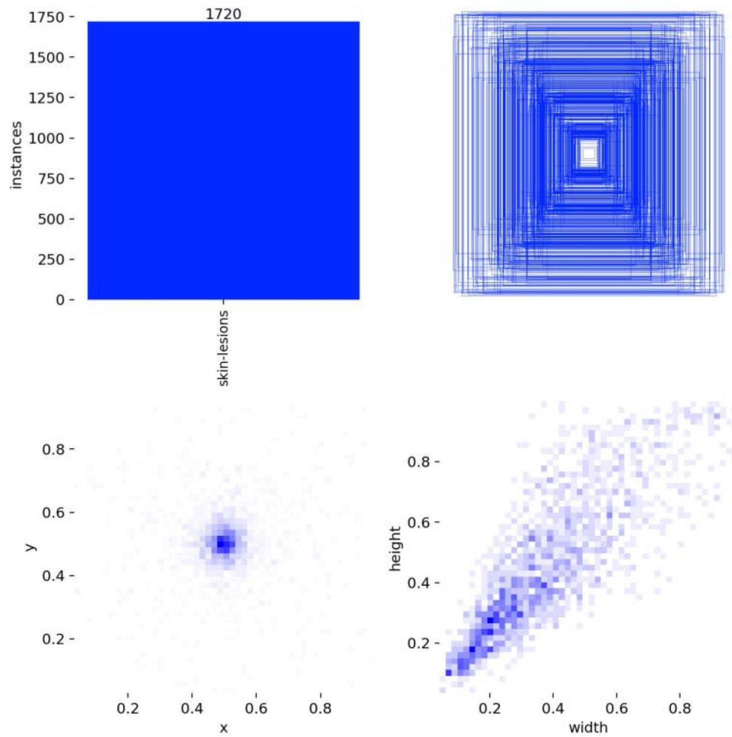


Figure 1: Distribution and annotation statistics of the dataset.

4. Methodology

The project was implemented using the YOLOv8 object detection model provided by the Ultralytics library. The overall workflow is shown below:

- Download the skin lesion dataset from Roboflow.
- Prepare the dataset in YOLO format.
- Train the YOLOv8 model using the training images.
- Validate the model after each epoch.
- Evaluate the trained model using standard metrics.
- Perform predictions on unseen skin lesion images.

YOLOv8 processes an image in a single forward pass and predicts the location of lesions using bounding boxes along with confidence scores. This makes the model both fast and suitable for dermatological image screening.

5. Training Configuration

The model was trained using the following configuration:

Parameter	Value
Model	YOLOv8n
Epochs	50
Image Size	640 × 640
Batch Size	16
Optimizer	Auto (Ultralytics)
Classes	1 (Skin Lesion)

The model was trained on annotated skin lesion images, where each lesion region was marked using bounding boxes.

6. Results and Discussion

The trained model showed good convergence during training and achieved strong detection performance on the validation dataset.

6.1 Training Performance

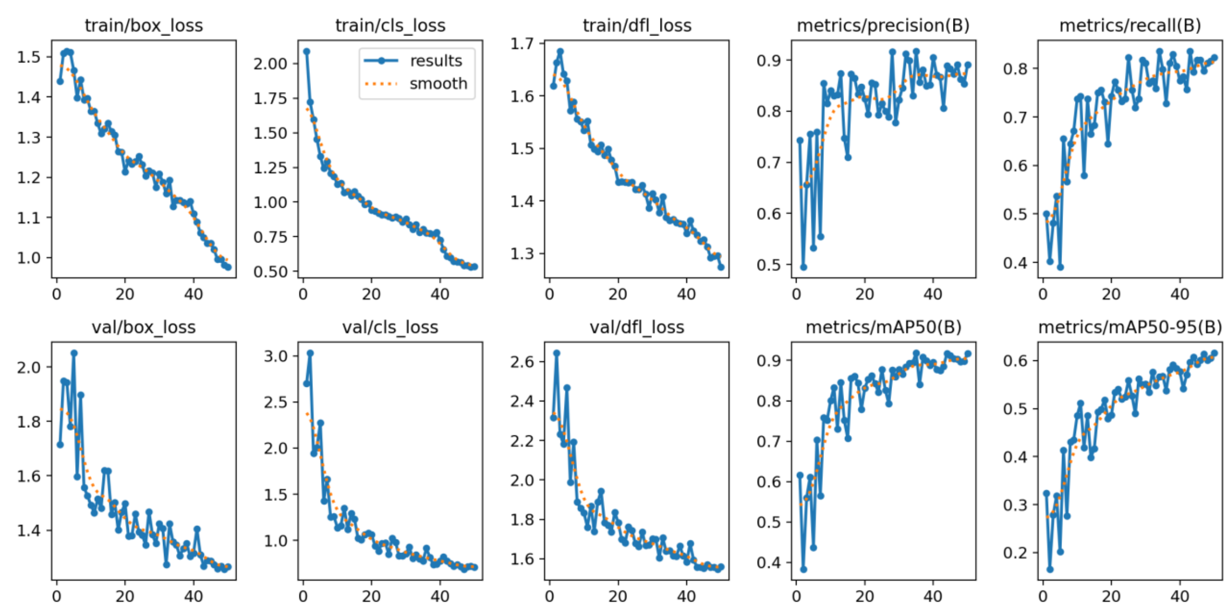


Figure 2: Training and validation loss, precision, recall, and mAP curves over 50 epochs.

The training and validation losses consistently decreased over the training epochs, indicating that the model successfully learned meaningful features from the skin lesion images. Precision, recall, and mAP values steadily improved during training, though some fluctuation is visible in the earlier epochs before the metrics stabilized after around epoch 20.

6.2 Performance Metrics

The final evaluation (epoch 50) produced the following results:

Metric	Value
Precision	0.89
Recall	0.82
F1 Score	0.85
mAP@0.5	0.92
mAP@0.5:0.95	0.62

These results indicate that the model is capable of detecting skin lesions with high accuracy at a moderate localization tolerance (mAP@0.5), while the stricter mAP@0.5:0.95 metric, which averages performance over tighter IoU thresholds, reflects the more challenging task of precisely outlining irregular lesion boundaries.

6.3 Precision-Recall Curve

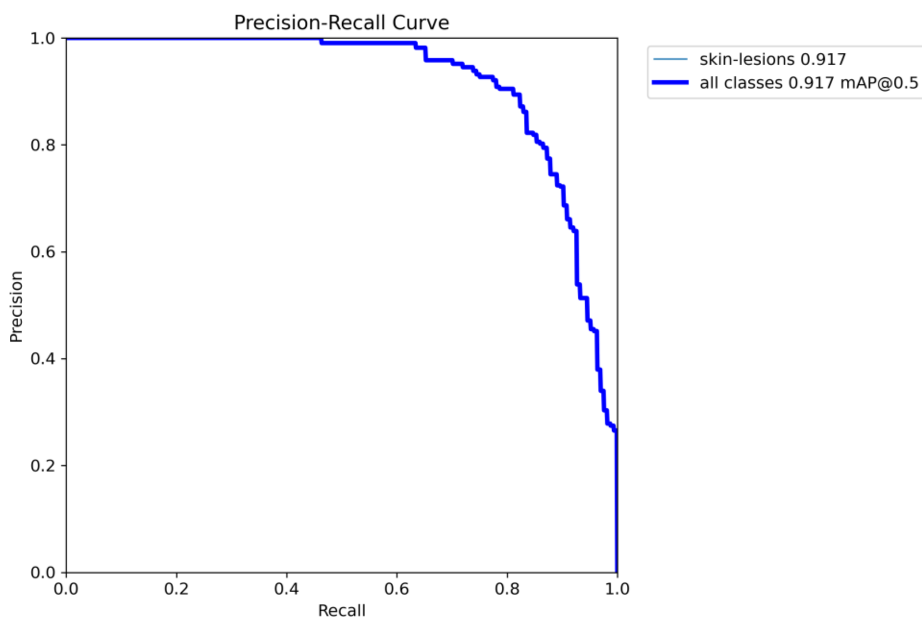


Figure 3: Precision-Recall curve — mAP@0.5 = 0.917.

The Precision-Recall curve demonstrates the relationship between precision and recall at different confidence thresholds. The curve stays close to the upper-right region for most of its range before dropping off at very high recall values, indicating strong overall detection performance with some difficulty in capturing the last few, likely more ambiguous, lesion instances.

6.4 F1-Confidence Curve

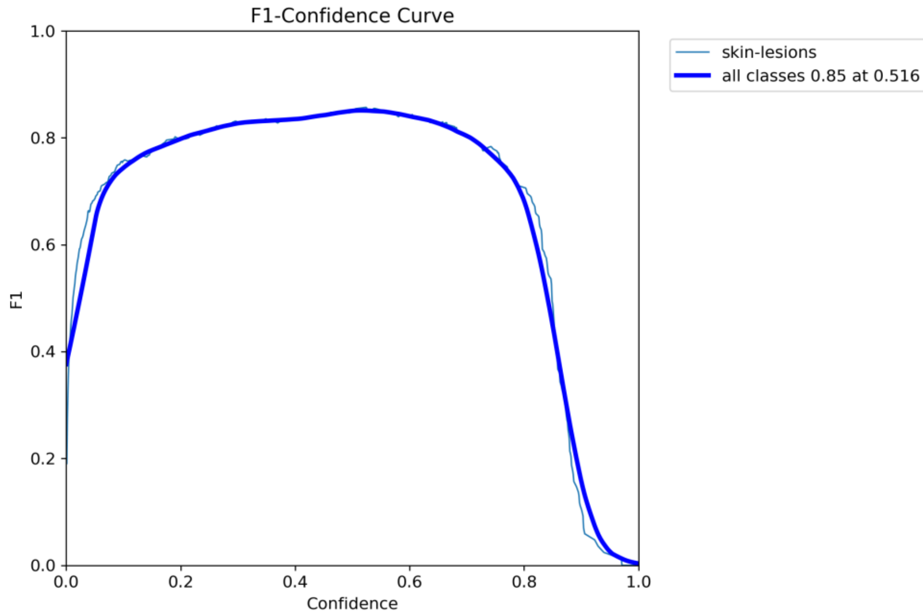
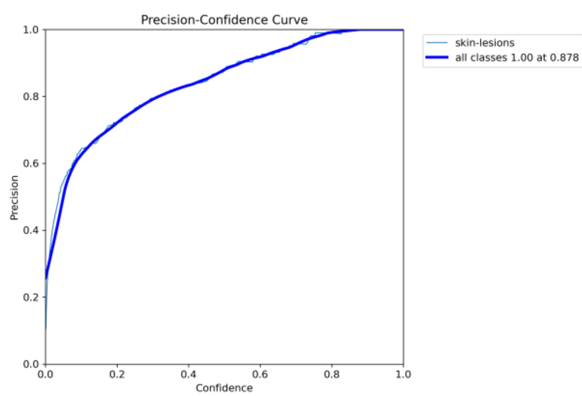


Figure 4: F1-Confidence curve — best F1 = 0.85 at confidence 0.516.

The F1-Confidence curve shows how the F1 score changes with different confidence thresholds. The model achieves its best F1 score of approximately 0.85 at a confidence threshold of 0.516, suggesting that an intermediate threshold provides the best balance between precision and recall for this task.

6.5 Precision and Recall Curves



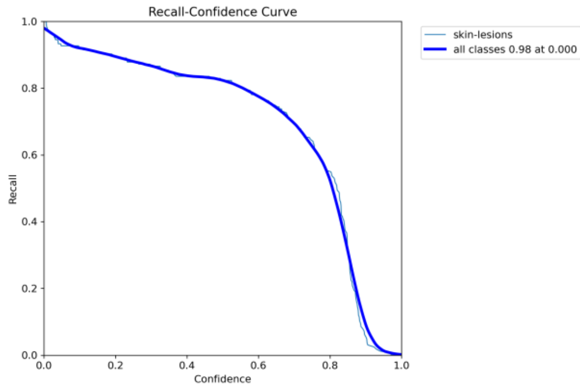


Figure 5: Precision-Confidence curve (left) and Recall-Confidence curve (right).

The Precision-Confidence curve shows that precision increases steadily as the confidence threshold becomes higher, reaching close to 1.0 at very high thresholds. The Recall-Confidence curve shows a gradual decline in recall as the threshold increases. This behaviour is expected because stricter confidence thresholds reduce false positives but may miss some true detections.

6.6 Confusion Matrix

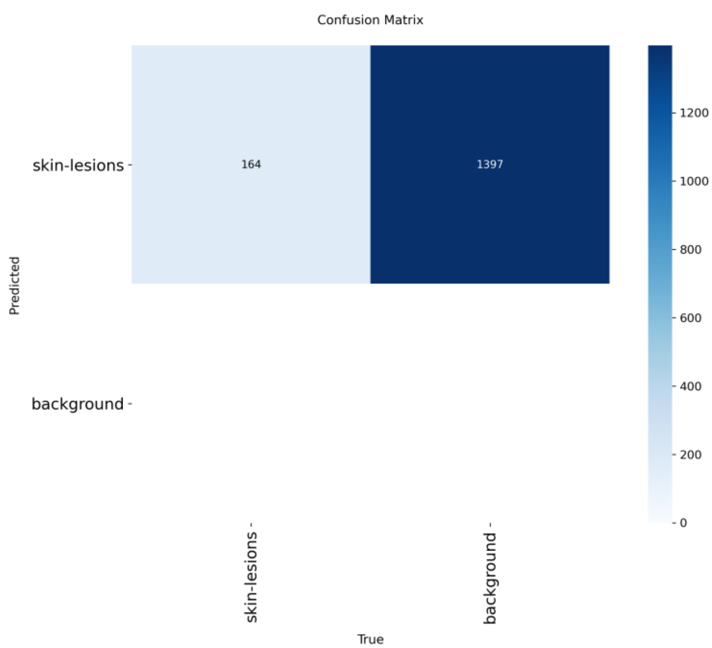


Figure 6: Confusion Matrix (raw counts, at default confidence threshold 0.25).

At the default confidence threshold of 0.25, the confusion matrix shows that 164 lesion instances were correctly matched to ground truth (true positives), while a large number of predicted boxes (1,397) did not correspond to any annotated lesion and were counted as false positives against the background class. No ground-truth lesions were left completely undetected at this low threshold.

This pattern — high recall but a high false-positive count at conf = 0.25 — is consistent with the F1-Confidence curve, which shows that raising the confidence threshold toward 0.5 substantially improves the balance between precision and recall by filtering out many of these low-confidence spurious detections.

6.7 Prediction Results

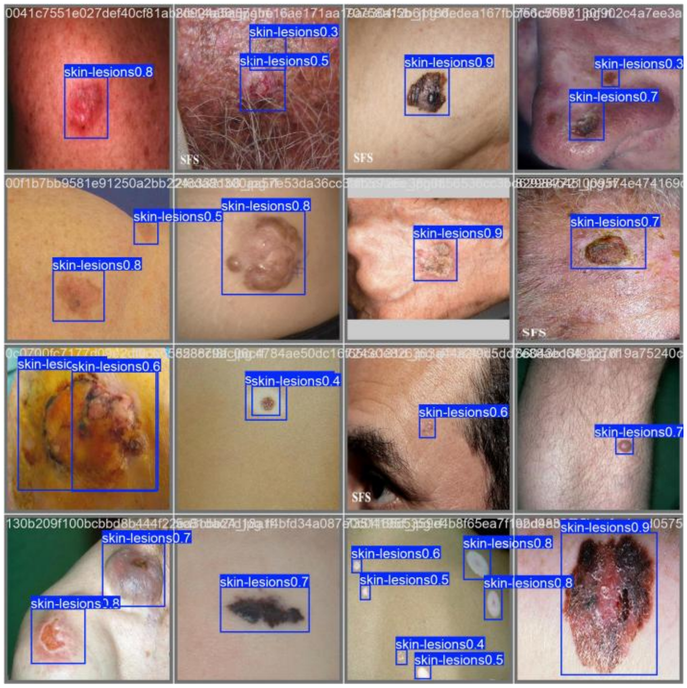


Figure 7a: Validation batch 0 predictions — bounding boxes with confidence scores overlaid on skin images.

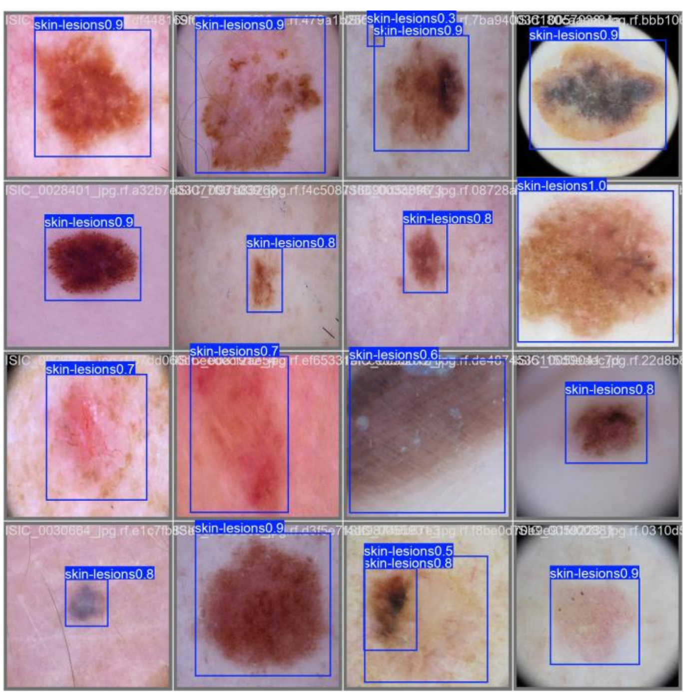


Figure 7b: Validation batch 1 predictions.

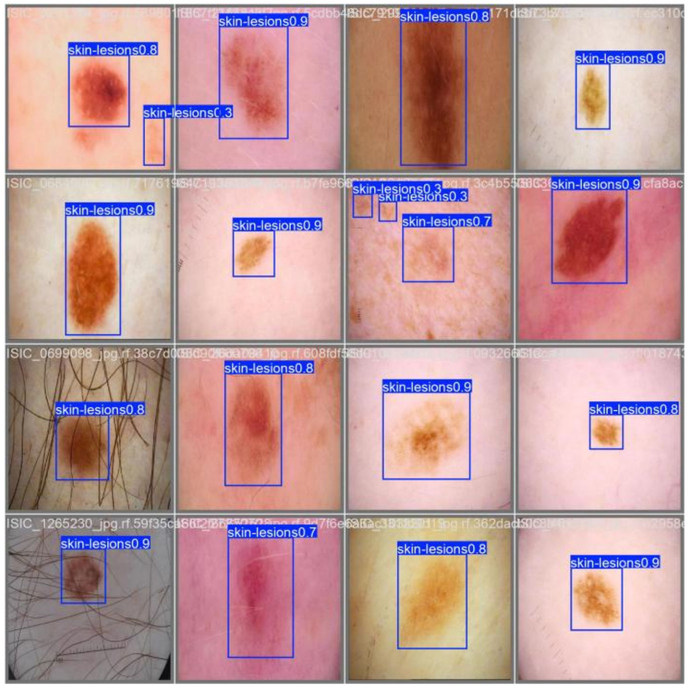


Figure 7c: Validation batch 2 predictions.

The validation predictions demonstrate that the trained model identifies skin lesion regions across a variety of image conditions, lighting, and skin tones. Most clearly visible lesions are enclosed within bounding boxes with reasonable confidence scores, though smaller or low-contrast lesions occasionally result in weaker or duplicate detections, consistent with the false-positive rate seen in the confusion matrix.

7. Conclusion

In this study, an automated framework for detecting skin lesions was developed by leveraging the YOLOv8 object detection architecture. The model, trained on curated dermoscopic/clinical image data, was able to identify and localize lesion regions with a reasonable degree of accuracy.

The evaluation metrics revealed solid performance, achieving an mAP@0.5 of 0.92, a precision of 0.89, and a recall of 0.82. Validation results confirmed that the system generates bounding boxes around lesions with useful confidence levels, although the elevated false-positive rate at low confidence thresholds indicates room for further tuning. Such findings suggest that YOLOv8 is a promising tool for assisting in the preliminary screening of skin lesions.

While this model is not a replacement for clinical expertise, it can serve as a supportive triage tool, helping flag regions of interest for closer dermatological examination.

8. Limitations

Several constraints were identified during this project:

- The training data was restricted to a single class (skin lesion), without distinguishing between benign and malignant types.
- Detection performance is tied to the resolution, lighting, and variety of the source images.
- A noticeably high false-positive rate was observed at low confidence thresholds, requiring careful threshold selection.
- The system provides localization only, lacking lesion classification (e.g., melanoma vs. benign nevus) or severity grading.

9. Future Work

Future iterations of this research could focus on:

- Utilizing larger, more heterogeneous skin image datasets for training.
- Adding multi-class support to differentiate between melanoma, benign nevi, and other lesion types.
- Enhancing robustness through advanced hyperparameter optimization, data augmentation, and false-positive suppression techniques.
- Developing a specialized web or mobile interface for real-time preliminary skin screening.

10. References

1. Jocher, G., et al. Ultralytics YOLOv8 Documentation.

2. Roboflow Universe. Skin Lesion Detection Dataset.

3. Redmon, J., et al. You Only Look Once: Unified, Real-Time Object Detection. CVPR, 2016.

4. Current literature on skin lesion and melanoma detection.